Practice workbook · Convert units and write numbers in standard form or significant figures

Unit conversions including length and time, rounding to significant figures, and writing values in standard form.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-NUMBER-FORM-UNITS

# 1. Question 3 · part a.ii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.ii only.

**3** (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

**(1)**

people with diabetes = .....

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

**(2)**

people with diabetes = .....

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

**(1)**

.....

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**10**

## 1. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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## 2. Question 3 · part b.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

**3** (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

**(2)**

| | **Steps** |
|---|---|
| | select a suitable objective lens |
| 5 | adjust focusing wheel to obtain a clear image |
| 1 | remove cells from the inside of the cheek |
| | place microscope slide onto the microscope stage |
| | smear the cell sample onto a microscope slide |

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

**(1)**

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

**(1)**

.....

.....

(iv) Give two ways that the student could avoid health hazards when working with living cells.

**(2)**

1 .....

.....

2 .....

.....

**6**

P 6 2 0 5 2 A 0 6 2 4

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## 2. Question 3 · part b.ii · 2 marks

November 2020 Paper 1R

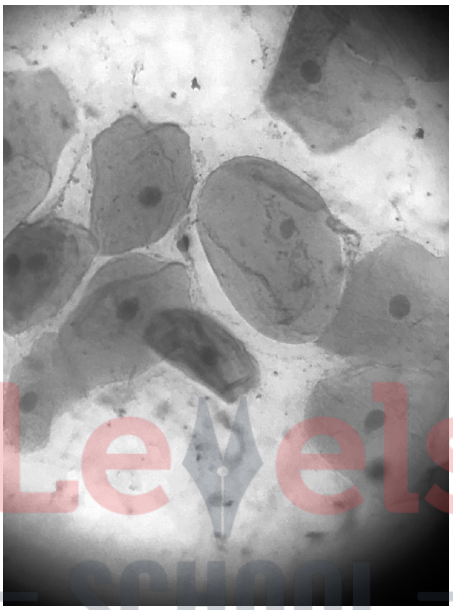
Further practice · Attempt part b.ii only.

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(b) The photograph shows an image of a cheek cell seen under a light microscope using a 10× eyepiece lens and a 40× objective lens.



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(i) Calculate the total magnification used to view the cheek cell.

**(1)**

total magnification = .....

(ii) The actual diameter of the cheek cell shown in the photograph is 0.018 mm.

Convert the diameter of the cheek cell into micrometre (μm).
[1 μm = $10^{-6}$ m]

**(2)**

diameter = ..... μm

**(Total for Question 3 = 9 marks)**

## 2. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 3. Question 3 · part a.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.i only.

**3** (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

**(1)**

people with diabetes = .....

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

**(2)**

people with diabetes = .....

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

**(1)**

.....

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## 3. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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# 4. Question 3 · part a.ii · 1 mark

January 2023 Paper 2

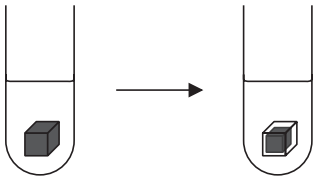
Further practice · Attempt part a.ii only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

.....

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## 4. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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